

Research Papers

Optimal charging of lithium-ion batteries based on lithium precipitation suppression

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ABSTRACT

With the development of the times, lithium-ion batteries have gradually become the main power source for new energy electric vehicles. The development of new energy electric vehicle charging technology cannot be separated from the progress of lithium-ion battery charging technology. Due to the secondary reaction of lithium precipitation during the lithium-ion battery charging process, there is a higher demand for lithium-ion battery charging strategies. This paper introduces an optimized charging method for suppressing lithium precipitation based on the second-order RC equivalent circuit model of lithium-ion batteries. An adaptability function is established to determine the lithium precipitation suppression value and temperature increase value, and the particle swarm algorithm is used to optimize the current value for each charging stage. The effectiveness of this charging method based on the particle swarm algorithm is verified through contrasted simulations with the method of suppressing lithium precipitation at 20 °C in MATLAB/Simulink, which proves its efficiency and superiority.

1. Introduction

With the rapid development of industrialization, mineral resources have been recklessly exploited and wasted, and the ecological environment has been seriously damaged and polluted. The development of new energy electric vehicles can effectively meet the current needs for resource conservation and environmental protection [1]. Therefore, the sales of new energy electric vehicles have seen explosive growth. According to relevant departments, as of the end of 2020, new energy buses have been promoted and used in 466,000 units nationwide, along with 132,000 new energy taxis [2]. According to statistics from a certain Chinese auto association, the number of new energy vehicles reached 6.887 million in 2022, an increase of 93.4 % compared with the previous year. However, the development of new energy electric vehicles is highly dependent on charging technology. Safe and fast charging technology can not only effectively improve the operating capacity of new energy electric vehicles, but also enhance their application scenarios.

Lithium-ion batteries, with advantages including their long lifespan, high temperature resistance, large capacity, small size, and lack of memory effects [3], have been widely used in new energy electric

vehicles, serving as the “heart” of these vehicles and providing a power source. Therefore, the charging methods for lithium-ion batteries have received great attention from many domestic and foreign scholars, who have proposed a series of charging methods, such as traditional charging methods [4], multi-stage charging methods [5,6], and algorithm-based charging methods [7]. While exploring safe and fast charging methods for lithium-ion batteries, scholars have analyzed the charging principles of lithium-ion batteries and found that the movement of lithium ions and electrons during charging leads to the risk of lithium precipitation if the charging method is not appropriate or done at low temperatures. This can further lead to the formation of dendrites and dead lithium [8], which not only consumes Li^+ in the electrolyte, affecting the battery capacity and shortening its lifespan, but may also puncture the separator, leading to thermal runaway and battery combustion or explosion.

In response to the safety issues caused by lithium precipitation during the battery charging process, this article proposes an optimized charging method for lithium-ion batteries that inhibits lithium precipitation. A fitness function is established based on the lithium inhibition value and temperature rise value, and a particle swarm algorithm is used to search for the optimal charging current values under various charging

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demands. The proposed method has been experimentally proven to effectively improve the charging efficiency and extend the battery life of new energy vehicles.

2. The working principle of the battery and lithium precipitation

2.1. The working principle of battery

In 1980, A. Armand first proposed that the essence of the charging and discharging reaction in lithium-ion batteries is the process of lithium ions going back and forth between the positive and negative electrodes to be de-embedded and embedded, and called it “rocking chair battery”. When the battery is charged, lithium ions will be generated at the positive electrode of the battery, and the generated lithium ions will move to the negative electrode through the electrolyte. The carbon (negative electrode) has a layered structure with many micro-pores in which the lithium ions transported from the positive electrode to the negative electrode can be embedded. The more lithium ions are intercalated, the higher the charging capacity. At the same time, the generation of lithium ions on the positive electrode causes the appearance of an electron, which moves from the positive to negative electrode through an external circuit. Similarly, during discharge (i.e., use of the battery), lithium ions intercalated in the negative electrode carbon layer deintercalated and move back to the positive electrode. The more lithium ions that return to the positive electrode, the higher the discharge capacity. Fig. 1 shows the working principle of a lithium-ion battery.

2.2. Side reactions during charging of lithium-ion batteries

2.2.1. Lithium precipitation

The internal process of lithium ion battery charging can be described as the lithium ions first separate from the oxide of the positive electrode and then pass through the diaphragm and electrolyte into the graphite micropores of the negative electrode, while the electrons move from the positive electrode to the negative electrode relying on peripheral circuitry. In the process of lithium ion movement, (i) if the rate of lithium ion departure from the positive electrode is too fast, which prevents the equal entry of lithium ions into the graphite micropores at the negative electrode; (ii) if there is too much resistance to the entry of lithium ions into the negative electrode; and (iii) if there is not enough space for lithium ion insertion in the negative electrode, those lithium ions that cannot enter the graphite can only stay on the surface of the negative

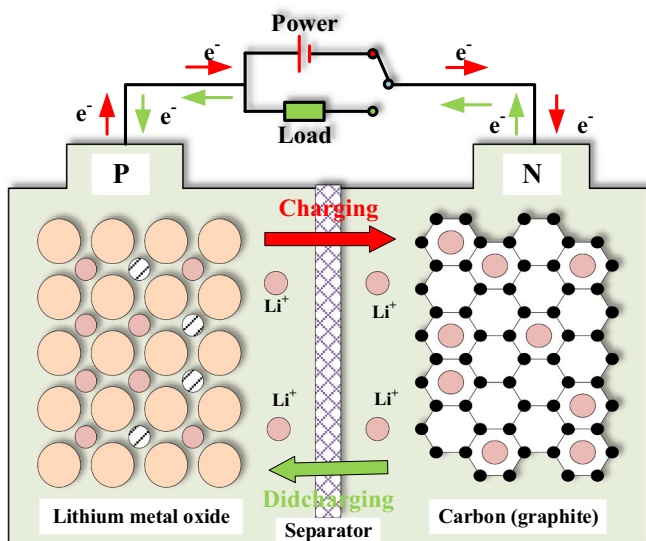


Fig. 1. Operating principle of lithium-ion battery.

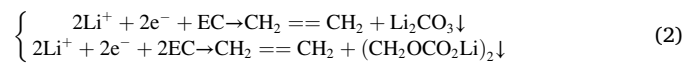
electrode, where they combine with the moving electrons to form a metallic lithium substance, in a phenomenon known as lithium precipitation [9], expressed as follows:



The process of lithium precipitation is illustrated in Fig. 2. Initially, lithium ions are reduced to metallic lithium on the surface of the negative electrode. Due to various factors, the deposited lithium is not uniformly distributed on the surface, and lithium dendrites are formed in the raised areas where it accumulates [8]. During charging, the dendrites grow in a straight line, with no significant changes in their shape, indicating that their growth starts from the point of contact with the negative electrode surface. Later, the dendrites become detached, with their front ends breaking away from the main body and disconnecting from the negative electrode, resulting in the formation of “dead lithium” [10]. The formation of lithium dendrites results in the consumption of Li^+ in the electrolyte, which directly affects the battery's capacity. Additionally, there is a risk of puncturing the separator, which can lead to internal short circuits, battery combustion or explosion. Moreover, the formation and continuous accumulation of “dead lithium” make it difficult for lithium ions to be transported through the negative electrode, as they encounter many “obstacles” on their path [11].

2.2.2. SEI film

The initial SEI film is formed during the first charge/discharge cycle of the battery, but it is still possible for the SEI film to form during subsequent charge/discharge cycles. This is mainly because SEI films are formed by a side reaction between the electrolyte (Ethylene Carbonate, EC), lithium ions and electrons, the expression of which is shown in Eq. (2). From the equation, it can be concluded that the main products Li_2CO_3 and $(\text{CH}_2\text{OCO}_2\text{Li})_2$ are the main components generated on the negative electrode surface to form SEI (Solid Electrolyte Interphase) film.



Both the SEI film and the separator have the function of allowing lithium ions to pass freely while restricting the passage of electrons. Therefore, the SEI film generated on the negative electrode surface does not hinder the insertion of lithium ions into the negative electrode and the charging of the battery; however, it impedes the passage of electrons, preventing side reactions from occurring. However, the metallic lithium deposited on the negative electrode surface has the opposite functionality of letting electrons pass freely while restricting the passage of lithium ions. Therefore, when electrons pass through the metallic lithium, further lithium dendrite growth may occur or secondary SEI film generation may take place via contact with lithium ions or EC in the electrolyte [12]. Fig. 3 illustrates different deposits on the local surface of the negative electrode, showing the paths of lithium ions, EC, and electrons, as well as the location where side reactions occur, indicated by the arrows.

Therefore, lithium-ion batteries in the charging process side reaction precipitation of lithium is more serious harm to the battery.

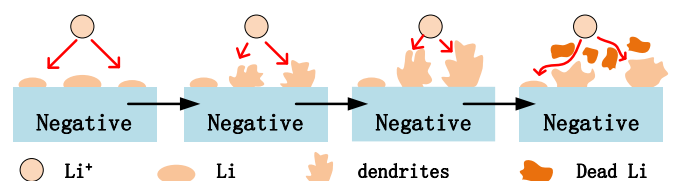


Fig. 2. Diagram of lithium precipitation change process.

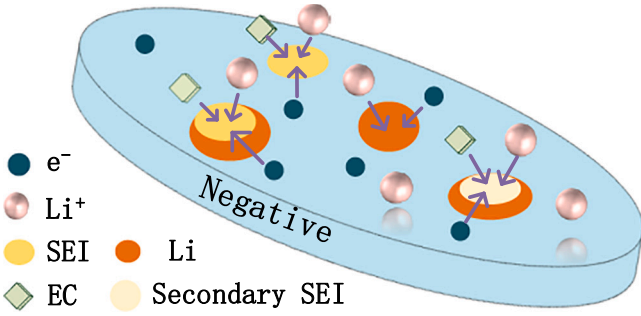


Fig. 3. Schematic diagram of different deposits on the local surface of the negative electrode.

2.3. Lithium precipitation judgment

During the charging process of lithium-ion batteries, deposition of lithium caused by side reactions can pose harm to the battery and affect its safety. To avoid lithium precipitation caused by side reactions, it is necessary to develop a scientifically reasonable charging method based on criteria for lithium precipitation in batteries. According to existing literature, criteria for lithium precipitation in batteries can be divided into saturation concentration criteria [13] and potential criteria for lithium precipitation [14–16]. Saturation concentration criteria consider that during the charging process of a lithium-ion battery, the rate at which lithium ions are embedded in the negative electrode surface is faster than the rate at which lithium ions diffuse into the negative electrode. Therefore, lithium ions gradually accumulate at the negative electrode surface. When the concentration of lithium ions exceeds the maximum allowable concentration at the negative electrode surface, lithium precipitation will occur as a side reaction. Potential criteria for lithium precipitation consider that lithium precipitation is an electrochemical reaction and can be limited by controlling the size of the overpotential for lithium precipitation. When the equilibrium potential for lithium precipitation in the battery (0 V) is greater than the difference between the solid-phase and liquid-phase potentials of the negative electrode of the battery, lithium precipitation will begin as a side reaction. In 2017, Haoge from Tsinghua University [17] conducted experiments to compare these criteria and found that the potential criteria for lithium precipitation were more accurate. As lithium precipitation is an electrochemical reaction, controlling the net current for lithium precipitation based on overpotential is more consistent with its essence. Therefore, this paper will use potential criteria for lithium precipitation to develop a scientifically reasonable charging method.

During the charging process of a lithium-ion battery, side reactions typically occur on the surface of the negative electrode. The overall reaction current is equal to the sum of the intercalation reaction current and the lithium precipitation reaction current. The expression for the lithium precipitation reaction current is given by Eq. (3) [18].

$$j_{Li-D} = a_s i_{0,Li-D} \left[\exp\left(\frac{\alpha_{a,Li-D} F \eta_{Li-D}}{RT}\right) - \exp\left(\frac{-\alpha_{c,Li-D} F \eta_{Li-D}}{RT}\right) \right], \quad \eta_{Li-D} \leq 0V \quad (3)$$

where a_s is the relative specific surface area, $i_{0,Li-D}$ is the side reaction lithium precipitation exchange current density, $\alpha_{a,Li-D}$ and $\alpha_{c,Li-D}$ are the electrochemical reaction transfer coefficients, η_{Li-D} is the lithium precipitation overpotential, R is the ideal gas constant, F is Faraday's constant and T is the temperature. The expression for the lithium precipitation overpotential is Eq. (4) [18]:

$$\eta_{Li-D} = \varphi_s - \varphi_l - U_{e,L} - U_{e,SEI} \quad (4)$$

where φ_s is the negative solid-phase potential, φ_l is the negative liquid-

phase potential, $U_{e,L}$ is the lithium precipitation equilibrium potential (generally regarded as 0 V), and $U_{e,SEI}$ is the potential pressure drop caused by the SEI film (which will be neglected).

According to Eq. (3), the lithium precipitation reaction occurs when $\eta_{Li-D} \leq 0V$, and according to Eq. (4), when Eq. (5) is satisfied, there will be no lithium precipitation when the lithium-ion battery is charged.

$$\eta_{Li-D} = \varphi_s - \varphi_l > 0 \quad (5)$$

For the lithium precipitation overpotential to satisfy the condition of Eq. (5) is inputted into the coupled thermal-electrochemical model of lithium-ion batteries, the suppressed lithium precipitation charging current-SOC line clusters can be obtained at different temperatures, as shown in Fig. 4. [16,17]. As shown in the figure, overall, the maximum allowed charging current for the battery gradually decreases throughout the entire charging process. During the early stage of charging when the SOC is low, the battery allows a larger maximum charging current with less likelihood of lithium precipitation reactions. However, as charging proceeds and SOC increases, the maximum allowed charging current gradually decreases and the lithium precipitation reactions can be easily triggered. Therefore, if the charging current of the lithium-ion battery exceeds the current limit for inhibiting lithium deposition, lithium precipitation reactions will occur, which overall poses a certain level of risk to the battery. Conversely, when charging current remains below this current limit, lithium precipitation reactions will not occur, and the charging process will be safe and stable, contributing to maintaining the battery's health. (Lithium precipitation and lithium deposition are synonymous).

3. Battery optimization charging strategy for inhibiting lithium dendrite formation

3.1. Lithium-ion battery model

3.1.1. Equivalent circuit model

In this paper, we will use the second-order RC model in the equivalent circuit model of lithium-ion battery for MATLAB/Simulink charging experimental simulation, and its model diagram is shown in Fig. 5. Where, U is the battery terminal voltage, U_{ocv} is the open circuit voltage (OCV) of the battery; R_0 is the ohmic internal resistance, describing the ohmic polarization of the battery during charging; R_1 is the electrochemical polarization resistance, C_1 is the electrochemical polarization capacitance, $R_1 C_1$ circuit is used to describe the electrochemical polarization of the battery during charging; R_2 is the

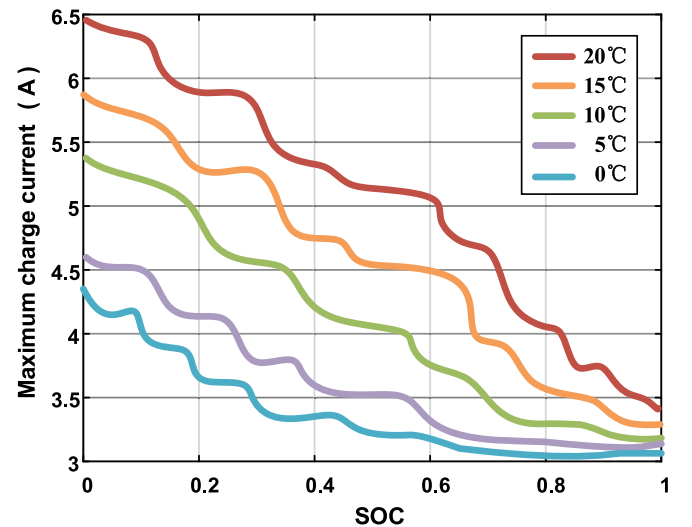


Fig. 4. Suppression of lithium precipitation charging current-SOC line clusters at different temperatures [16,17].

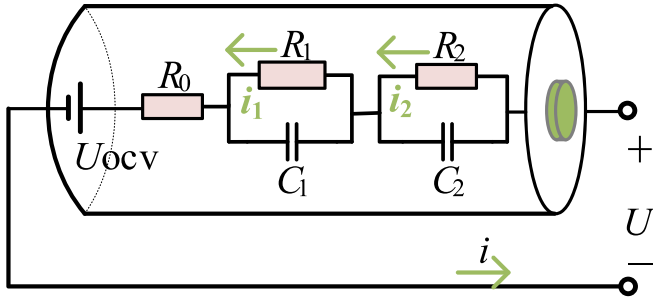


Fig. 5. Second-order RC equivalent circuit model.

concentration difference polarization resistance, C_2 is the concentration difference polarization capacitance, and the R_2C_2 loop is used to describe the concentration difference polarization during battery charging; i is the battery charging current, and i_1 and i_2 are the currents flowing through R_1 and R_2 . The RC parameters in its model are referred to the literature [19].

3.1.2. Temperature model

During the charging process of a battery, the heat generated by the internal chemical reactions and the internal resistance of the battery itself lead to a certain temperature rise in the battery. The temperature rise model is established based on the second-order RC model mentioned above. Assuming that the internal temperature of the battery is uniform and selecting the average internal temperature of the battery as the battery temperature, the heat equation of a uniformly exchanged object with the environment can be solved to calculate the battery temperature, expressed as follows:

$$C_T \frac{dT}{dt} = P_l - \frac{T - T_a}{R_T} \quad (6)$$

where C_T is the heat capacity, P_l is the power loss, R_T is the convective resistance, T_a is the ambient temperature, and T is the cell temperature. The Laplace transformation of the equation and the expression is as follows:

$$T(s) = \frac{P_l R_T + T_a}{1 + R_T C_T s} \quad (7)$$

3.2. The charging method for inhibiting lithium precipitation at 20 °C

Maximizing the charging current curve for lithium-ion batteries at 20 °C with suppressed lithium precipitation was obtained based on the lithium-ion battery model. Fig. 4 shows the clusters of charging current-SOC curves with suppressed lithium precipitation at different temperatures, which provided the maximum charging current curve for the temperature of 20 °C. As the charging current curve is complex, a multi-stage constant current charging method with a stair-like charging strategy was utilized in this study. The charging current was divided into different stages with 10 % SOC intervals, and a 2 % safety margin was added to each stage to ensure that the constant charging current value for the overall charging process remained below the maximum charging current curve of the battery. Specifically, the stair-like charging current for each stage is shown in Fig. 6.

On the basis of a Li-ion battery model, a Li-ion battery charging simulation experiment was conducted using the multi-stage constant current charging current presented in Fig. 6 at 20 °C. Based on the simulation experimental results, the charging time required for inhibiting lithium precipitation at 20 °C (20 °C-ILP) is approximately 3072 s, with a maximum temperature increase of approximately 3.3 °C.

However, the maximum charging current curve at 20 °C was obtained under isothermal conditions, and it was found through the experiment that the maximum temperature rise of 3.3 °C no longer met

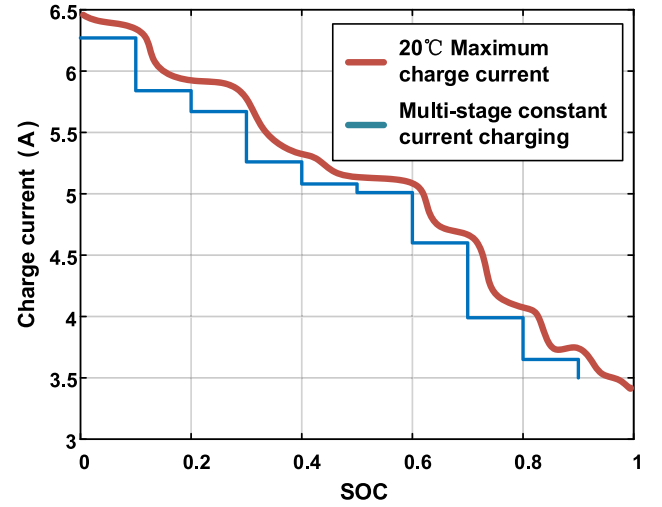


Fig. 6. Multi-stage constant current charging current at 20 °C.

the isothermal conditions. According to related research [17], the maximum charging current that suppresses Li precipitation will vary with temperature changes. If the temperature of the Li-ion battery rises during charging, the maximum charging current that suppresses Li precipitation will also increase, as shown in Fig. 7 [17]. As shown in Fig. 7, under adiabatic conditions at 20 °C, the maximum charging current curve for the Li-ion battery changes with the increase of the battery temperature, and in comparison, with the current curve under isothermal conditions at 20 °C, the difference in the maximum charging current in the early charging stage is not significant, whereas difference becomes significant in the middle and later stages of charging, and the charging current under adiabatic conditions clearly increases.

3.3. Particle swarm optimization charging study considering lithium deposition suppression

3.3.1. Lithium suppression model

Based on the phenomenon of temperature rise during lithium-ion battery charging, this article proposes a maximum charging current that varies with temperature during the charging process of the battery, allowing lithium-ion batteries to suppress lithium deposition and shorten charging time. It is assumed that curve l_2 is a charging current curve in the battery charging model, while curve l_1 is the lithium

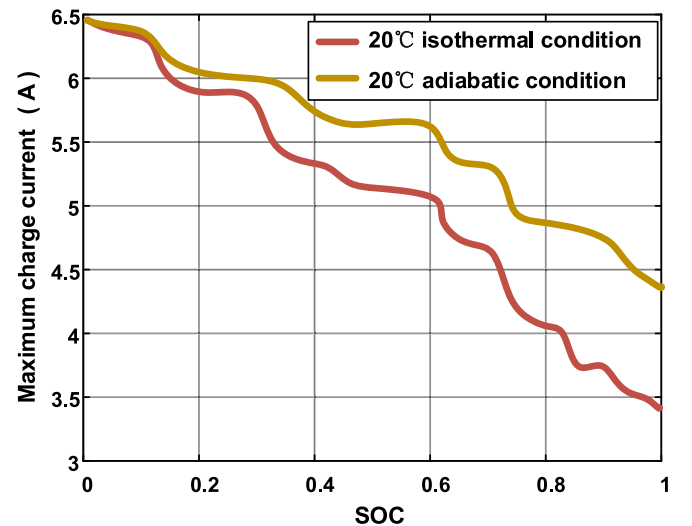


Fig. 7. Maximum charge current to suppress lithium precipitation [17].

deposition suppression charging current curve that varies with temperature under adiabatic conditions at 20 °C in Fig. 7. The area between I_1 and I_2 is defined as the lithium deposition suppression region, denoted as I_{YL} , as shown in Fig. 8.

Calculation of the lithium precipitation inhibition region leads to the following equation for the lithium inhibition value:

$$I_{YL}(t) = \int \{I_1(t) - I_2(t)\} dt \quad (8)$$

Discretizing Eq. (8) by taking the step size Δt , then:

$$\begin{cases} I_1[k] = I_1[k \times \Delta t] \\ I_2[k] = I_2[k \times \Delta t] \end{cases} \quad (9)$$

Considering that the charging current value remains constant from $k \times \Delta t$ to $(k + 1) \times \Delta t$ time, the expression of the discretized I_{YL} , i.e., the suppressed lithium model, is Eq. (10):

$$\begin{cases} W_{YL}[k] = \sum \{I_2[k] - I_1[k]\} \Delta t \\ W_{YL}[k + 1] = \sum \{I_2[k + 1] - I_1[k + 1]\} \Delta t \end{cases} \quad (10)$$

When the charging current of the lithium-ion battery, I_2 , approaches or reaches I_1 , the lithium deposition suppression region is smaller, indicating a lower suppression value for lithium deposition. However, the battery temperature rise will be greater, i.e., a higher temperature rise value. Conversely, if the lithium deposition suppression value is larger, the temperature rise value will be smaller. Therefore, to reconcile the relationship between the suppression value for lithium deposition and the temperature rise value, this paper will use a particle swarm algorithm to design a fitness function to solve for the charging current value.

3.3.2. Optimization of charging method with particle swarm algorithm accounting for lithium dendrite suppression

In order to simulate the charging process of lithium-ion batteries, it is necessary to set the initial charging conditions, charging current magnitude, and charging cutoff conditions. This information required for battery charging is then integrated into a particle swarm optimization algorithm. The specific steps of this method are as follows:

- Initialization: The charging start condition is set to SOC 0, and the population size n is set to 20 with the maximum number of iterations M set to 30. The learning factors c_1 and c_2 are set to 1.5. The particle's velocity and position ranges are also set, along with the initial best fitness values and positions for individuals and populations. The

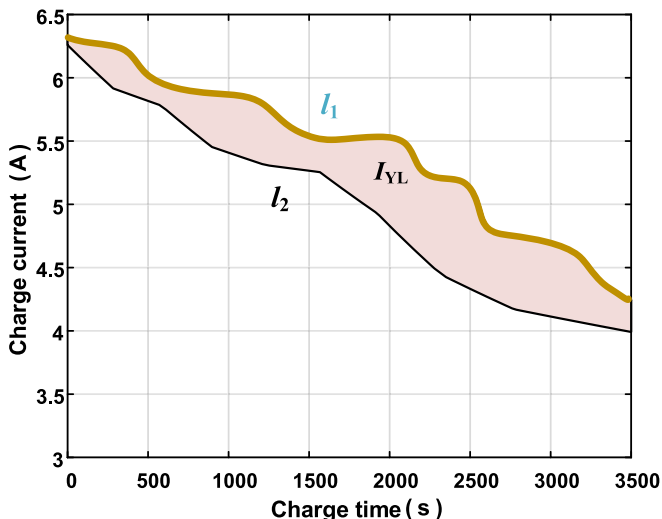


Fig. 8. Suppress lithium precipitation region.

position range of particles is set to the maximum charging current I_1 that varies depending on the temperature during the charging process. Suitable charging current values are selected based on the principle of step multi-stage constant current charging.

- Fitness function: In order to distinguish between the quality of evolution of each particle, the fitness value of each particle is calculated based on the fitness function. By establishing a temperature model and a lithium inhibition model for lithium-ion batteries, the temperature rise value and the lithium inhibition value can be obtained through segmented charging. Therefore, we use the temperature rise value and lithium inhibition value of the battery during charging as the evaluation criteria for the fitness function, expressed mathematically as follows:

$$W_{WS} = \int (T - T_a) dt \quad (11)$$

$$\begin{cases} F = \alpha \times \frac{W_{YL}}{W_{YL-ref}} + \beta \times \frac{W_{WS}}{W_{WS-ref}} \\ \alpha + \beta = 1 \end{cases} \quad (12)$$

where, α is the weight coefficient of lithium suppression, β is the weight coefficient of temperature rise, and the corresponding weight coefficients of lithium suppression and temperature rise can be set according to different needs; W_{YL} is the lithium suppression value of battery charging, W_{YL-ref} is the lithium suppression value of 1C-CC charging method; W_{WS} is the temperature rise of battery charging, W_{WS-ref} is the temperature rise value of 1C-CC charging method. The fitness value of each particle is calculated according to Eq. (12).

- Optimization: The individual optimal fitness value (each particle compares its current fitness value with its own previous fitness value and selects the smaller value as the individual optimal fitness value for each particle) and individual optimal position are searched for based on the fitness values of each particle. The population optimal fitness value (the smallest fitness value from among the particles is selected as the population optimal fitness value) and corresponding position are then searched for based on the individual optimal fitness values. This is done in order to find the optimal combination of charging current for lithium-ion batteries.
- Updating: The velocity and position update formulas for the particles are shown in Eqs. (13) and (14):

$$V_i(t+1) = w \times V_i(t) + c_1 \times rand \times [pBest_i(t) - X_i(t)] + c_2 \times rand \times [gBest(t) - X_i(t)] \quad (13)$$

$$X_i(t+1) = X_i(t) + V_i(t+1) \quad (14)$$

where $V_i(t)$ is the velocity of the i th particle in the t th iteration and $V_i(t+1)$ is the velocity of the i th particle in the $t+1$ th iteration; c_1 is the individual learning factor, $rand$ is a random number between [0,1], $pBest_i(t)$ is the individual optimal position of the i th particle in the t th iteration, and $X_i(t)$ is the position of the i th particle in the t th iteration; c_2 is the population learning factor, $gBest(t)$ is the population optimal position of the particle in the t th iteration; w is the inertia weight, and in order to obtain the optimal result, the time-varying inertia weight is set as follows:

$$w = w_{max} - (w_{max} - w_{min}) \times \frac{t}{M} \quad (15)$$

where, w_{max} is the maximum value of inertia weight, usually taken as 0.9; w_{min} is the minimum value of inertia weight, usually taken as 0.4; t is the number of iterations; M is the maximum number of iterations.

- Boundary handling for velocity and position: For updated particles, the range of their velocity and position is limited to the values set during initialization. If the particle's velocity or position exceeds the

maximum value, their values are reset to the maximum. Similarly, if the particle's velocity or position falls below the minimum value, their values are reset to the minimum.

- **Iteration:** Return to step 2) and continue running until the termination condition is met. In this study, the termination condition for one iteration is when the state of charge (SOC) reaches 90 %. The termination condition for the entire algorithm is when the maximum number of iterations is reached. The flowchart of the algorithm based on particle swarm optimization for optimizing the charging process is shown in Fig. 9.

4. Simulation and experiments

4.1. Simulation and analysis

4.1.1. Setting optimisation targets

Based on the lithium-ion battery model, temperature model and lithium inhibition model built in MATLAB/Simulink, and according to the fitness function established in the particle swarm algorithm, this paper sets up a charging method that changes the ratio of weight coefficients to achieve different optimization objectives. The ratio of the first weighting coefficient is the ratio of the weighting coefficients of the lithium inhibition value to the temperature rise value is $\alpha:\beta = 9:1$, which represents that lithium-ion batteries take the inhibition of lithium precipitation as the main optimization objective during charging, and the reduction of battery temperature rise as the secondary optimization objective. The second has a weight ratio of $\alpha:\beta = 7:3$, while the third has a weight ratio of $\alpha:\beta = 5:5$, which means that the lithium-ion battery takes into account both suppressing lithium dendrite growth and reducing battery temperature rise during charging. The fourth has a weight ratio of $\alpha:\beta = 3:7$, and the fifth has a weight ratio of $\alpha:\beta = 1:9$, which indicates that reducing battery temperature rise is the primary optimization goal during charging.

The above five different optimization objectives are sequentially substituted into the computation of the fitness function of the particle swarm algorithm, so that the cell calculates the fitness values under different optimization objectives according to the fitness function formula during the optimization process.

4.1.2. Simulation analysis

Take the first weight coefficient ratio of $\alpha:\beta = 9:1$ as an example to describe the process of charging optimization of lithium-ion battery:

- The initialization in the particle swarm algorithm was used to determine the size of the 20 particles (the size of the charging current);
- The particles are transferred one by one to the battery model, which starts charging from the transferred particles, i.e. the charging current; when the battery is fully charged, the lithium suppression and temperature rise values obtained from the simulation are immediately transferred back to the algorithm.
- In the algorithm, the fitness value of a particle is calculated from the fitness function Eqs. (11) and (12);
- In the algorithm, according to the fitness value of this particle, the comparison of its own fitness value is carried out to find the smaller fitness value as the individual optimal fitness value; secondly, the comparison of fitness value between particles is carried out and the smaller fitness value is the group optimal fitness value;
- In the algorithm, this particle is updated according to the update Eqs. (13)–(15) to bring it closer to the neighborhood of the population optimal particle;
- In the algorithm, velocity and position boundaries are applied to the particles;
- The updated particles are again transferred to the battery model for battery charging simulation, i.e., returning to the second step, running and cycling sequentially until the cut-off condition is met to end the simulation; finally, the data recorded by the algorithm can be used to obtain the group-optimal particles (optimal charging current values) that the particles have found during the cycling process.

Taking the ratio of the first and second weight coefficients as an example, the iterative curve of its particles to find the optimal fitness value is shown in Fig. 10. Among them, the particles in the optimization process of the algorithm is also in the process of constantly looking for a smaller fitness value, in which one particle cycles 30 times, making 20 particles cycle a total of 600 times. And because the ratio of the weight coefficients obtained is different, resulting in different values of the

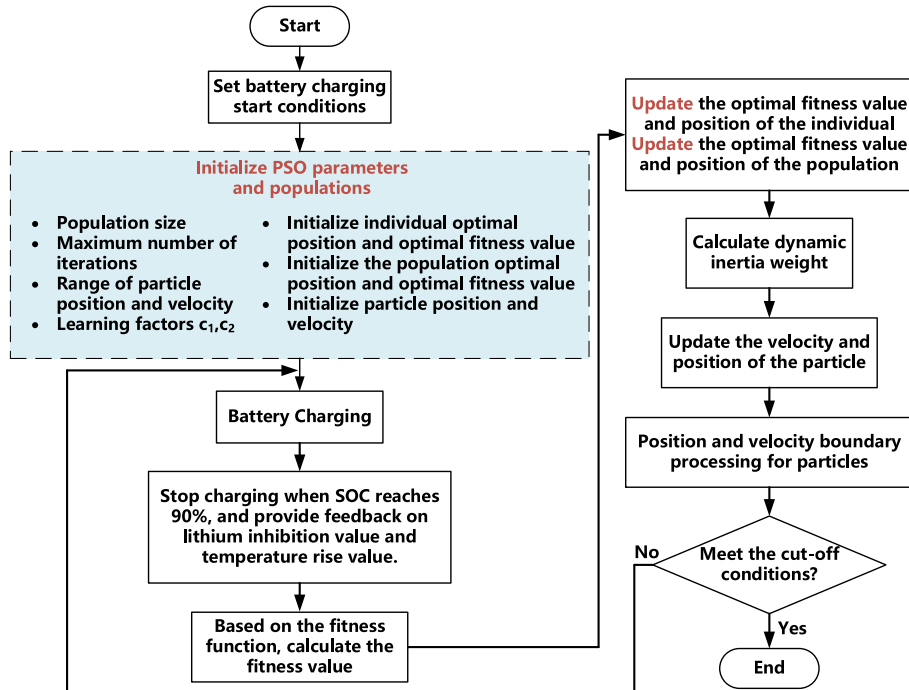
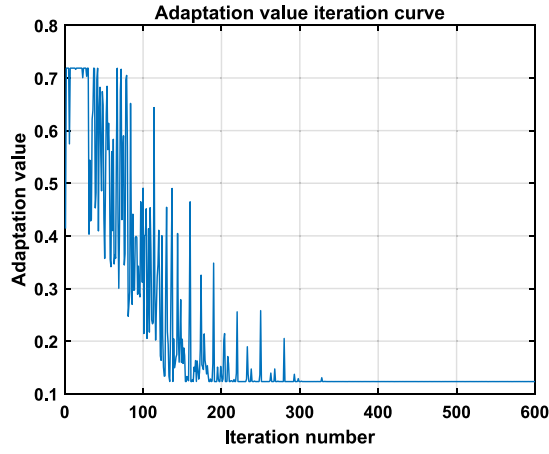
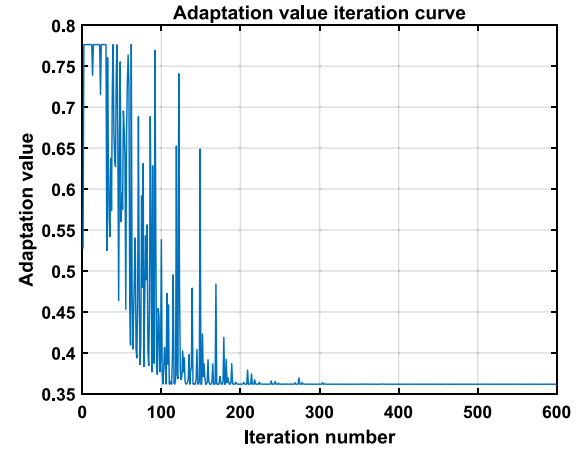


Fig. 9. Battery charging based on particle swarm optimization algorithm.



(a)



(b)

Fig. 10. Adaptation value iteration curve: (a) $\alpha:\beta = 9:1$; (b) $\alpha:\beta = 7:3$.

fitness function in the calculation because of different ratios, it can be seen from Fig. 10 that when the lithium suppression value accounts for a larger proportion, the optimal fitness value is smaller; and Fig. 10 also shows that the fitness value of the iterative curve shows a gradual convergence of the tendency to converge to a stable value about the time of the cycle of about 300 times, which proves that the particles in the algorithmic optimization process to achieve the goal of finding the optimal particle and achieve the optimization of the particle. Particles in the optimization process of the algorithm, achieving the optimization effect.

4.2. Experiments and analyses

4.2.1. Experimental setup

This article reports on a particle swarm algorithm optimized charging experiment for suppressing lithium precipitation through an in-situ hardware experimentation platform, as illustrated by Fig. 11. The hardware experimentation platform comprises a host machine, real-time target machine, oscilloscope, and charging circuitry. The real-time target machine utilized in the experiment is Speedgoat, which provides hardware, basic IO boards, driver libraries, and FPGA programming software modules. The experiment demonstrates the successful

suppression of lithium precipitation during charging, thereby improving the safety and performance of lithium-ion batteries.

4.2.2. Experimental analysis

In this section, the optimal particles (optimal charging current values) searched by the particle swarm algorithm under different optimization objectives in the previous Section 4.1 are used to conduct charging experiments on lithium-ion batteries, and the data under different optimization objectives can be obtained, as shown in Table 1.

As can be seen from Table 1, as the weighting factor α of the lithium suppression value gradually decreases, the weighting factor β of the

Table 1
Simulation data under different optimization objectives.

Optimization goals	W_{YL}	W_{WS}	Charging time (s)	Maximum temperature rise ($^{\circ}\text{C}$)
$\alpha:\beta = 9:1$	12.9	5436.8	2701	3.7
$\alpha:\beta = 7:3$	18.4	5417.4	2703	3.6
$\alpha:\beta = 5:5$	482.5	4664.7	2800	3.3
$\alpha:\beta = 3:7$	1496.4	4325.9	2994	2.9
$\alpha:\beta = 1:9$	2326.5	4223.8	3170	2.7
20 $^{\circ}\text{C}$ -ILP	–	–	3072	3.3

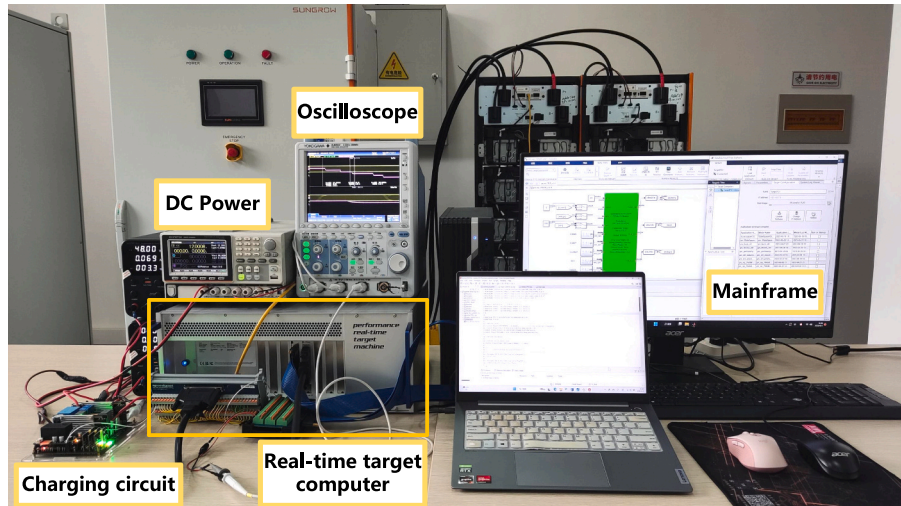


Fig. 11. Hardware-in-the-loop experimental platform physical diagram.

temperature rise gradually increases, resulting in the lithium suppression value increasing while the temperature rise value decreases. The charging time required under the first two optimization objectives ($\alpha:\beta = 9:1$ and $\alpha:\beta = 7:3$) is shorter than the charging time of 3072 s for the lithium precipitation inhibition charging method (20 °C-ILP) at 20 °C, but the maximum temperature rise is higher than 3.3 °C. The maximum temperature rise obtained under the third optimization objective ($\alpha:\beta = 5:5$) is the same as that of the 20 °C-ILP, but the charging is faster, while the fourth optimization objective ($\alpha:\beta = 3:7$) not only charges faster but also has a lower temperature rise compared to the 20 °C-ILP. Therefore, all the five charging methods with different optimization objectives set in this paper achieve their respective objectives.

Figs. 12 and 13 show the charging current-SOC plots and temperature profiles for five different optimization objectives (different weighting factor ratios) and 20 °C-ILP charging methods obtained by applying the optimal charging current values obtained in the previous Section 4.1 to the experimental setup. Among them, it can be obtained from Fig. 12, regardless of which charging method charging current is lower than the maximum charging current curve under the adiabatic condition of 20 °C to achieve the condition of lithium suppression; with the gradual decrease of the weight coefficient α of the lithium suppression value, the lithium suppression value is gradually increased that is, the battery charging current curve as a whole is moving away from the maximum charging current curve under the adiabatic condition of 20 °C, and the charging current is gradually shifted downward. From Fig. 13, as the temperature rise weighting factor β gradually increases, the temperature rise value is gradually decreasing, the battery temperature curve is gradually moving downward, and the maximum temperature rise value is also decreasing. In Fig. 13, it can be clearly seen that the charging method with the fourth optimization objective $\alpha:\beta = 3:7$ not only has a shorter charging time, but also has a lower maximum temperature rise compared to the 20 °C-ILP.

Based on Figs. 12 and 13, it can be seen that the lithium precipitation suppression charging method at 20 °C can be compared to the five different optimized charging methods set forth in this paper. The results show that the charging times of the first four methods are faster than that of the 20 °C-ILP method, with little difference in maximum temperature rise. Even the 3:7 $\alpha:\beta$ charging method not only has a shorter charging time but also has a lower maximum temperature rise. Therefore, it can be concluded that the present paper proposes a method based on the maximum charging current of a battery, which varies with the temperature during the charging process, to be used as a boundary value. Particle swarm optimization algorithm can be used to achieve the desired charging goals based on different charging requirements. This

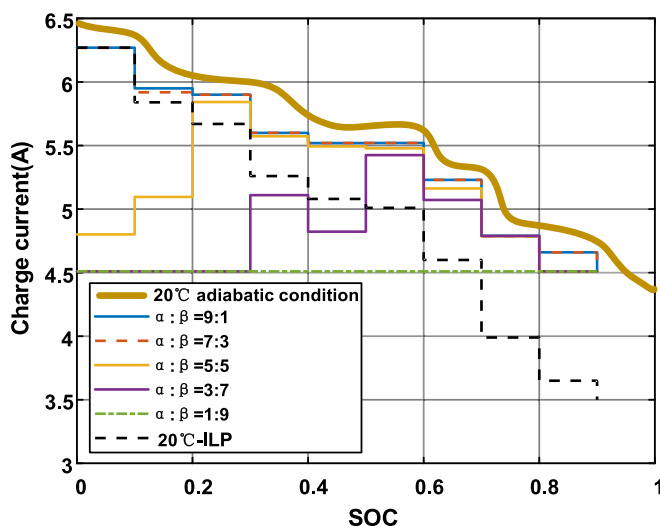


Fig. 12. Charging current-SOC diagram for different optimization targets.

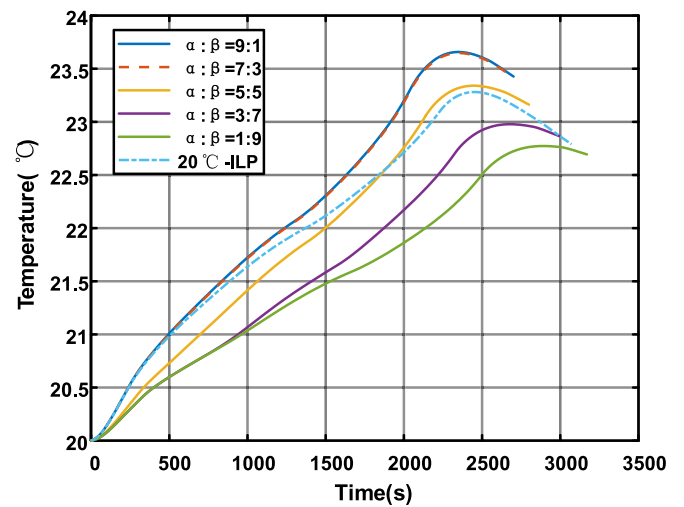


Fig. 13. Battery temperature diagram for different optimization targets.

method provides an efficient and effective way to optimize the charging process.

5. Conclusion

This article presents an optimized charging method for lithium-ion batteries based on the second-order RC equivalent circuit model to prevent lithium precipitation. The fitness function is established between the lithium precipitation suppression coefficient and temperature rise, and the particle swarm optimization algorithm is used to obtain the current value for each charging stage. Lithium-ion battery charging experiments were conducted on a hardware-environment experimental platform. From the experimental results, it can be concluded that the method proposed in this paper achieves different charging objectives, and the method with the optimization objective of $\alpha:\beta = 3:7$ is better than the 20 °C-ILP charging method in terms of charging time and temperature rise, which verifies the validity of the method proposed in this paper.

CRedit authorship contribution statement

Author 1 (First Author): Conceptualization, Methodology, Formal analysis, Funding acquisition;

Author 2: Investigation, Software, Data curation, Data analysis, Writing - original draft, Writing - review & editing;

Author 3: Investigation, Software, Data ANALYSIS, Writing - review & editing;

Author 4: Investigation, Software, Resources, Data Curation, Review;

Author 5: Investigation, Software, Review;

Author 6: Visualization, Writing - review & editing.

Declaration of competing interest

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Data availability

No data was used for the research described in the article.

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